

INTRODUCTION

The CE6275 series is a set of three-terminal low power high voltage regulator implemented in CMOS technology. They can deliver 150mA output current and allow an input voltage as high as 30V. They are available with several fixed output voltages ranging from 2.8V to 5V. CMOS technology ensures low voltage drop and low quiescent current.

Although designed primarily as fixed voltage regulators, these devices can be used with external components to obtain variable voltages and currents.

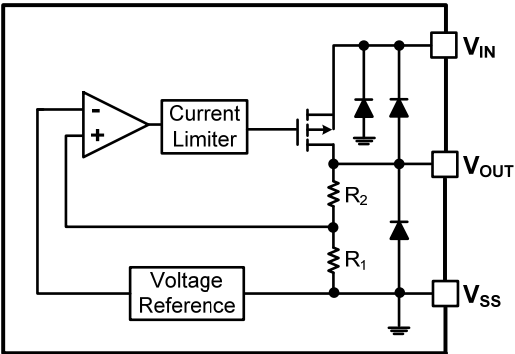
FEATURES

- Low Power Consumption
- Low Voltage Drop
- Output Voltage Range: 2.8V ~ 5V
- Low Temperature Coefficient
- High Input Voltage (up to 30V)
- Quiescent current: 2.5μA
- High Output Current :150mA(Typ.)
- Output Voltage Accuracy: ±2%/±1%
- Ceramic Capacitor Compatible
- SOT23-3/5, SOT89-3 and TO-92 Packages

APPLICATIONS

- Battery-powered Equipment
- Audio/Video Equipment
- Portable Consumer Equipment
- Communication Equipment

BLOCK DIAGRAM



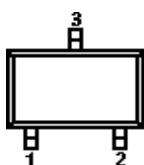
ORDER INFORMATIO

CE6275①②③④⑤

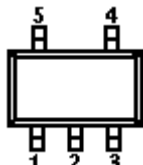
DESIGNATOR	SYMBOL	DESCRIPTION
①	A	Standard
②③	Integer	Output Voltage(2.8~5V) e.g:3.0V=②:3.③:0
④	M/MC/MY	Package:SOT23-3
	MR	Package:SOT23-5
	P	Package:SOT89-3
	T	Package:TO-92
⑤	-	Output Voltage Accuracy: ±2%
	1	Output Voltage Accuracy: ±1%

■ **PIN CONFIGURATION** (Pin output sequence can be ordered by customer)

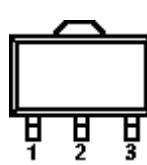
SOT23-3



SOT23-5



SOT89-3



TO-92



PIN NUMBER						PIN NAME	FUNCTION
SOT23-3			SOT89-3		TO-92		
M	MC	MY	P	PT	T		
1	3	3	1	2	1	V _{SS}	Ground
2	2	1	3	1	3	V _{OUT}	Output
3	1	2	2	3	2	V _{IN}	Power input

SOT23-5(MR)

PIN NUMBER	PIN NAME	FUNCTION
1	V _{SS}	Ground
2	V _{IN}	Power Input Pin
3	V _{OUT}	Output Pin
4,5	NC	No Connection

■ **Absolute Maximum Ratings**⁽¹⁾

(Unless otherwise specified, T_A=25°C)

PARAMETER	SYMBOL	RATINGS	UNITS
Input Voltage ⁽²⁾	V _{IN}	-0.3~33	V
Output Voltage ⁽²⁾	V _{OUT}	-0.3~10	V
Operating free air temperature range	T _A	-40~85	°C
Operating Junction Temperature range	T _J	-40~125	°C
Storage Temperature	T _{stg}	-40~125	°C
Lead Temperature(Soldering, 10 sec)	T _{solder}	260	°C

(1) Stresses beyond those listed under absolute maximum ratings may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the specification is not implied. Exposure to absolute-maximum-rated conditions for extended periods may affect device reliability.

(2) All voltages are with respect to network ground terminal.

■ Thermal Information

PARAMETER	SYMBOL	PACKAGE	MAX.	UNIT
Thermal Resistance (Junction to Ambient) (Assume no ambient airflow, no heats ink)	θ_{JA}	SOT23-3/5	500	°C/W
		SOT89-3	200	°C/W
		TO-92	200	°C/W
Power Dissipation	P_D	SOT23-3/5	400	mW
		SOT89-3	600	mW
		TO-92	500	mW

(1) P_D is measured at $T_A=25^\circ\text{C}$

■ Electrical Characteristics

CE6275A30+3.0V Output Type (Unless otherwise specified: $T_A=25^\circ\text{C}$)

PARAMETER	SYMBOL	TEST CONDITIONS	MIN.	TYP.	MAX.	UNIT
Input Voltage	V_{IN}	—	—	—	30	V
Output Voltage	$V_{OUT} (\pm 2\%)$	$V_{IN}=5.0\text{V}, I_{OUT}=10\text{mA}$	2.940	3.000	3.060	V
	$V_{OUT} (\pm 1\%)$		2.970	3.000	3.030	
Output Current	I_{OUT}	$V_{IN}=5.0\text{V}$	70	100	—	mA
Load Regulation	ΔV_{OUT}	$V_{IN}=5.0\text{V}, 1\text{mA} \leq I_{OUT} \leq 50\text{mA}$	—	5	25	mV
Dropout Voltage ⁽¹⁾	V_{DIF}	$I_{OUT}=1\text{mA}, \Delta V_{OUT}=2\%$	—	10	30	mV
Quiescent Current	I_{SS}	$V_{IN}=5.0\text{V}$, No load	—	2.5	4.0	μA
Line Regulation	$\frac{\Delta V_{OUT}}{\Delta V_{IN} \times V_{OUT}}$	$4.0\text{V} \leq V_{IN} \leq 30\text{V}, I_{OUT}=1\text{mA}$	—	0.1	0.2	%/V
Temperature Coefficient	$\frac{\Delta V_{OUT}}{\Delta T_A}$	$I_{OUT}=10\text{mA}, -40^\circ\text{C} < T_A < 85^\circ\text{C}$	—	± 0.45	—	mV/°C
Short Current	I_{SHORT}	$V_{OUT} = V_{SS}$	—	10	—	mA

(1) Dropout voltage is defined as the input voltage minus the output voltage that produces a 2% change in the output voltage from the value at $V_{IN} = V_{OUT}+2\text{V}$ with a fixed load.

CE6275A33+3.3V Output Type (Unless otherwise specified: $T_A=25^\circ\text{C}$)

PARAMETER	SYMBOL	TEST CONDITIONS	MIN.	TYP.	MAX.	UNIT
Input Voltage	V_{IN}	—	—	—	30	V
Output Voltage	$V_{OUT} (\pm 2\%)$	$V_{IN}=5.3\text{V}, I_{OUT}=10\text{mA}$	3.234	3.300	3.366	V
	$V_{OUT} (\pm 1\%)$		3.267	3.300	3.333	
Output Current	I_{OUT}	$V_{IN}=5.3\text{V}$	100	150	—	mA
Load Regulation	ΔV_{OUT}	$V_{IN}=5.3\text{V}, 1\text{mA} \leq I_{OUT} \leq 50\text{mA}$	—	5	25	mV
Dropout Voltage ⁽¹⁾	V_{DIF}	$I_{OUT}=1\text{mA}, \Delta V_{OUT}=2\%$	—	10	30	mV
Quiescent Current	I_{SS}	$V_{IN}=5.3\text{V}$, No load	—	2.5	4.0	μA
Line Regulation	$\frac{\Delta V_{OUT}}{\Delta V_{IN} \times V_{OUT}}$	$4.3\text{V} \leq V_{IN} \leq 30\text{V}, I_{OUT}=1\text{mA}$	—	0.1	0.2	%/V
Temperature Coefficient	$\frac{\Delta V_{OUT}}{\Delta T_A}$	$I_{OUT}=10\text{mA}, -40^\circ\text{C} < T_A < 85^\circ\text{C}$	—	± 0.5	—	mV/°C
Short Current	I_{SHORT}	$V_{OUT} = V_{SS}$	—	10	—	mA

(1) Dropout voltage is defined as the input voltage minus the output voltage that produces a 2% change in the output voltage from the value at $V_{IN} = V_{OUT}+2\text{V}$ with a fixed load.

■ Electrical Characteristics(continued)

CE6275A36+3.6V Output Type (Unless otherwise specified: $T_A=25^{\circ}\text{C}$)

PARAMETER	SYMBOL	TEST CONDITIONS	MIN.	TYP.	MAX.	UNIT
Input Voltage	V_{IN}	—	—	—	30	V
Output Voltage	$V_{OUT} (\pm 2\%)$	$V_{IN}=5.6\text{V}, I_{OUT}=10\text{mA}$	3.528	3.600	3.672	V
	$V_{OUT} (\pm 1\%)$		3.564	3.600	3.636	
Output Current	I_{OUT}	$V_{IN}=5.6\text{V}$	100	150	—	mA
Load Regulation	ΔV_{OUT}	$V_{IN}=5.6\text{V}, 1\text{mA} \leq I_{OUT} \leq 50\text{mA}$		5	25	mV
Dropout Voltage ⁽¹⁾	V_{DIF}	$I_{OUT}=1\text{mA}, \Delta V_{OUT}=2\%$	—	10	30	mV
Quiescent Current	I_{SS}	$V_{IN}=5.6\text{V}$, No load		2.5	4.0	μA
Line Regulation	$\frac{\Delta V_{OUT}}{\Delta V_{IN} \times V_{OUT}}$	$4.6\text{V} \leq V_{IN} \leq 30\text{V}, I_{OUT}=1\text{mA}$	—	0.1	0.2	%/V
Temperature Coefficient	$\frac{\Delta V_{OUT}}{\Delta T_A}$	$I_{OUT}=10\text{mA}, -40^{\circ}\text{C} < T_A < 85^{\circ}\text{C}$		± 0.6		mV/ $^{\circ}\text{C}$
Short Current	I_{SHORT}	$V_{OUT} = V_{SS}$		10		mA

(1) Dropout voltage is defined as the input voltage minus the output voltage that produces a 2% change in the output voltage from the value at $V_{IN} = V_{OUT}+2\text{V}$ with a fixed load.

CE6275A40+4.0V Output Type (Unless otherwise specified: $T_A=25^{\circ}\text{C}$)

PARAMETER	SYMBOL	TEST CONDITIONS	MIN.	TYP.	MAX.	UNIT
Input Voltage	V_{IN}	—	—	—	30	V
Output Voltage	$V_{OUT} (\pm 2\%)$	$V_{IN}=6.0\text{V}, I_{OUT}=10\text{mA}$	3.920	4.000	4.080	V
	$V_{OUT} (\pm 1\%)$		3.960	4.000	4.040	
Output Current	I_{OUT}	$V_{IN}=6.0\text{V}$	100	150	—	mA
Load Regulation	ΔV_{OUT}	$V_{IN}=6.0\text{V}, 1\text{mA} \leq I_{OUT} \leq 50\text{mA}$		5	25	mV
Dropout Voltage ⁽¹⁾	V_{DIF}	$I_{OUT}=1\text{mA}, \Delta V_{OUT}=2\%$	—	10	30	mV
Quiescent Current	I_{SS}	$V_{IN}=6.0\text{V}$, No load		2.5	4.0	μA
Line Regulation	$\frac{\Delta V_{OUT}}{\Delta V_{IN} \times V_{OUT}}$	$5.0\text{V} \leq V_{IN} \leq 30\text{V}, I_{OUT}=1\text{mA}$	—	0.1	0.2	%/V
Temperature Coefficient	$\frac{\Delta V_{OUT}}{\Delta T_A}$	$I_{OUT}=10\text{mA}, -40^{\circ}\text{C} < T_A < 85^{\circ}\text{C}$		± 0.7		mV/ $^{\circ}\text{C}$
Short Current	I_{SHORT}	$V_{OUT} = V_{SS}$		10		mA

(1) Dropout voltage is defined as the input voltage minus the output voltage that produces a 2% change in the output voltage from the value at $V_{IN} = V_{OUT}+2\text{V}$ with a fixed load.

■ Electrical Characteristics(continued)

CE6275A44+4.4V Output Type (Unless otherwise specified: $T_A=25^{\circ}\text{C}$)

PARAMETER	SYMBOL	TEST CONDITIONS	MIN.	TYP.	MAX.	UNIT
Input Voltage	V_{IN}	—	—	—	30	V
Output Voltage	$V_{OUT} (\pm 2\%)$	$V_{IN}=6.4\text{V}, I_{OUT}=10\text{mA}$	4.312	4.400	4.488	V
	$V_{OUT} (\pm 1\%)$		4.356	4.400	4.444	
Output Current	I_{OUT}	$V_{IN}=6.4\text{V}$	100	150	—	mA
Load Regulation	ΔV_{OUT}	$V_{IN}=6.4\text{V}, 1\text{mA} \leq I_{OUT} \leq 50\text{mA}$		5	25	mV
Dropout Voltage ⁽¹⁾	V_{DIF}	$I_{OUT}=1\text{mA}, \Delta V_{OUT}=2\%$	—	10	30	mV
Quiescent Current	I_{SS}	$V_{IN}=6.4\text{V}$, No load		2.5	4.0	μA
Line Regulation	$\frac{\Delta V_{OUT}}{\Delta V_{IN} \times V_{OUT}}$	$5.4\text{V} \leq V_{IN} \leq 30\text{V}, I_{OUT}=1\text{mA}$	—	0.1	0.2	%/V
Temperature Coefficient	$\frac{\Delta V_{OUT}}{\Delta T_A}$	$I_{OUT}=10\text{mA}, -40^{\circ}\text{C} < T_A < 85^{\circ}\text{C}$		± 0.7		$\text{mV}/^{\circ}\text{C}$
Short Current	I_{SHORT}	$V_{OUT} = V_{SS}$		10		mA

(1) Dropout voltage is defined as the input voltage minus the output voltage that produces a 2% change in the output voltage from the value at $V_{IN} = V_{OUT}+2\text{V}$ with a fixed load.

CE6275A50+5.0V Output Type (Unless otherwise specified: $T_A=25^{\circ}\text{C}$)

PARAMETER	SYMBOL	TEST CONDITIONS	MIN.	TYP.	MAX.	UNIT
Input Voltage	V_{IN}	—	—	—	30	V
Output Voltage	$V_{OUT} (\pm 2\%)$	$V_{IN}=7\text{V}, I_{OUT}=10\text{mA}$	4.900	5.000	5.100	V
	$V_{OUT} (\pm 1\%)$		4.95	5.000	5.05	
Output Current	I_{OUT}	$V_{IN}=7\text{V}$	100	150	—	mA
Load Regulation	ΔV_{OUT}	$V_{IN}=7\text{V}, 1\text{mA} \leq I_{OUT} \leq 50\text{mA}$		5	25	mV
Dropout Voltage ⁽¹⁾	V_{DIF}	$I_{OUT}=1\text{mA}, \Delta V_{OUT}=2\%$	—	10	30	mV
Quiescent Current	I_{SS}	$V_{IN}=7\text{V}$, No load		2.5	4.0	μA
Line Regulation	$\frac{\Delta V_{OUT}}{\Delta V_{IN} \times V_{OUT}}$	$6.0\text{V} \leq V_{IN} \leq 30\text{V}, I_{OUT}=1\text{mA}$	—	0.1	0.2	%/V
Temperature Coefficient	$\frac{\Delta V_{OUT}}{\Delta T_A}$	$I_{OUT}=10\text{mA}, -40^{\circ}\text{C} < T_A < 85^{\circ}\text{C}$		± 0.75		$\text{mV}/^{\circ}\text{C}$
Short Current	I_{SHORT}	$V_{OUT} = V_{SS}$		10		mA

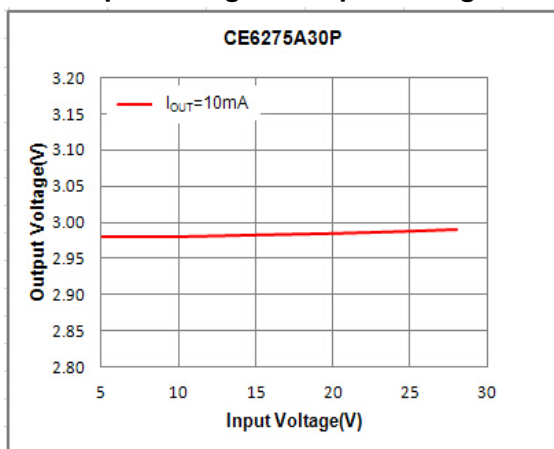
(1) Dropout voltage is defined as the input voltage minus the output voltage that produces a 2% change in the output voltage from the value at $V_{IN} = V_{OUT}+2\text{V}$ with a fixed load.

■ Typical Performance Characteristics

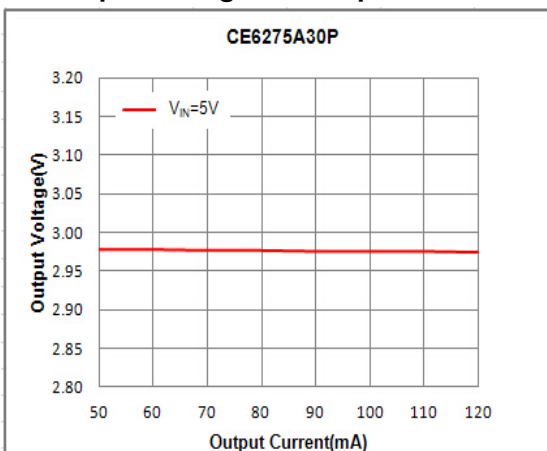
Test Condition: $V_{IN}=V_{OUT}+2V$, $T_A=25^{\circ}C$, unless otherwise specified

CE6275A30P Typ

Output Voltage Vs. Input Voltage

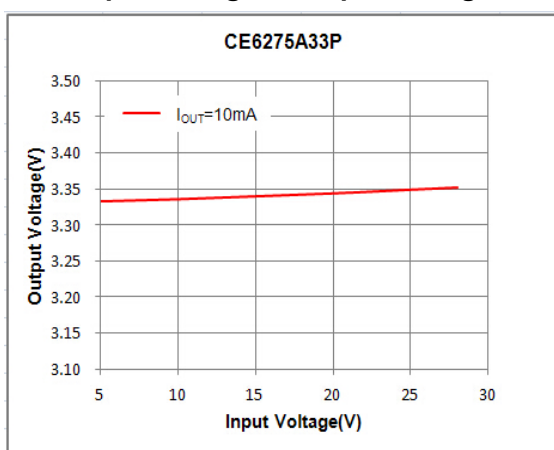


Output Voltage Vs. Output Current

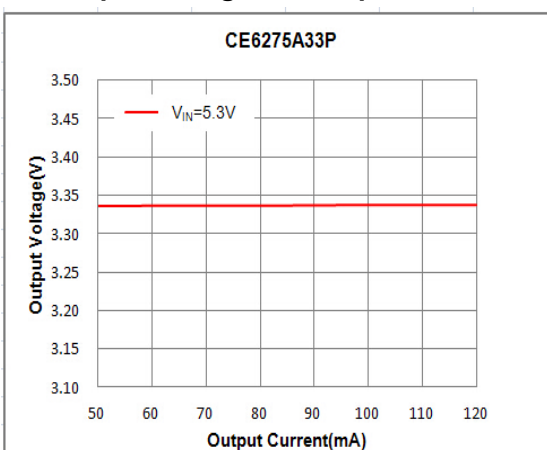


CE6275A33P Typ

Output Voltage Vs. Input Voltage

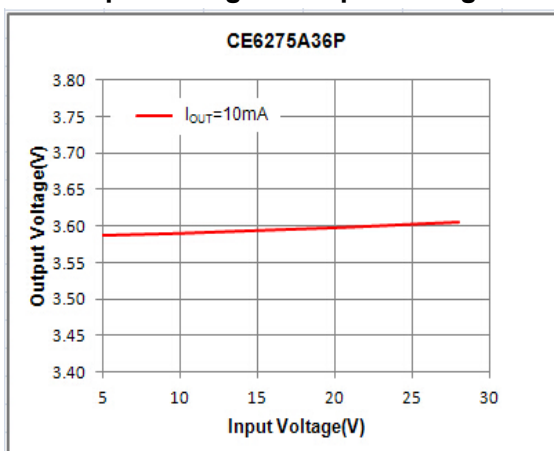


Output Voltage Vs. Output Current

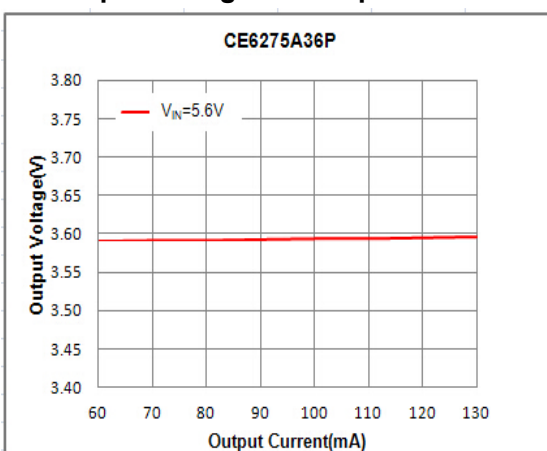


CE6275A36P Typ

Output Voltage Vs. Input Voltage



Output Voltage Vs. Output Current

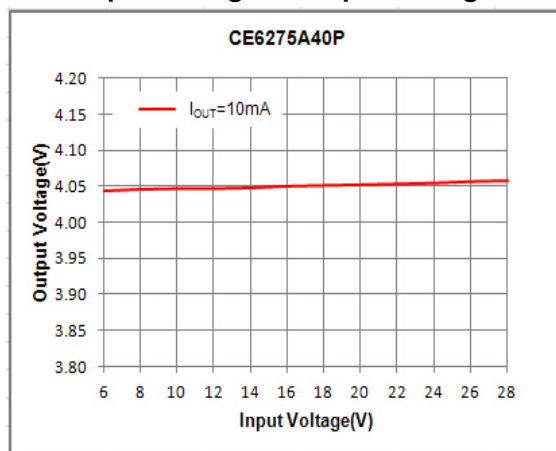


■ Typical Performance Characteristics(continued)

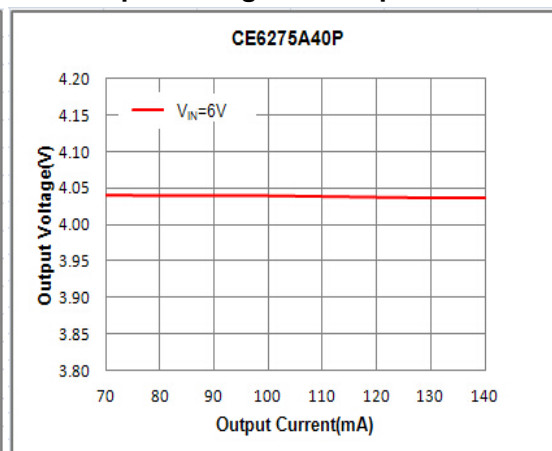
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CE6275A40P Typ

Output Voltage Vs. Input Voltage

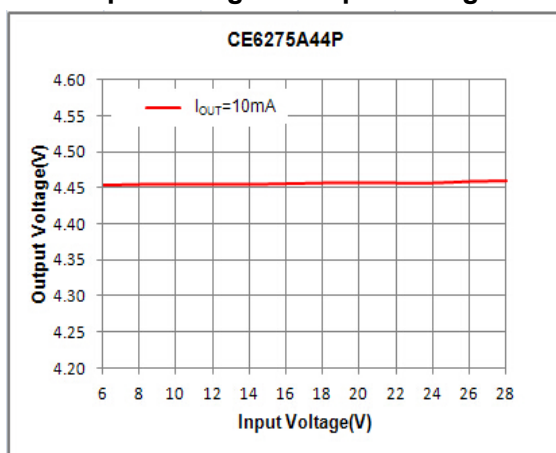


Output Voltage Vs. Output Current

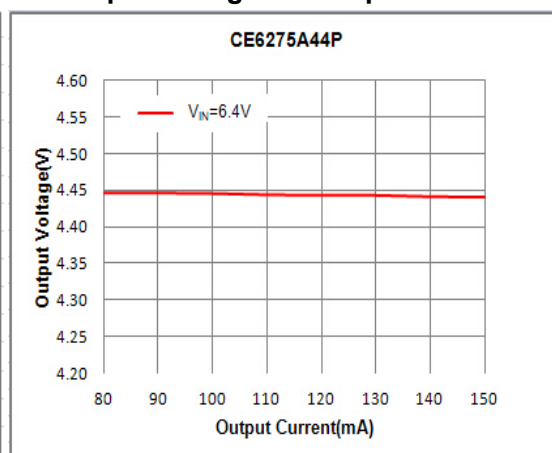


CE6275A44P Typ

Output Voltage Vs. Input Voltage

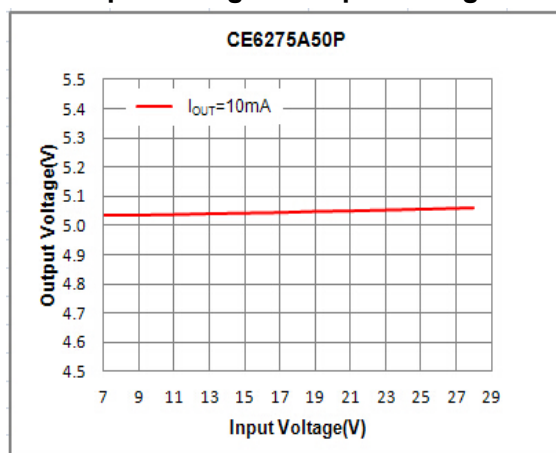


Output Voltage Vs. Output Current

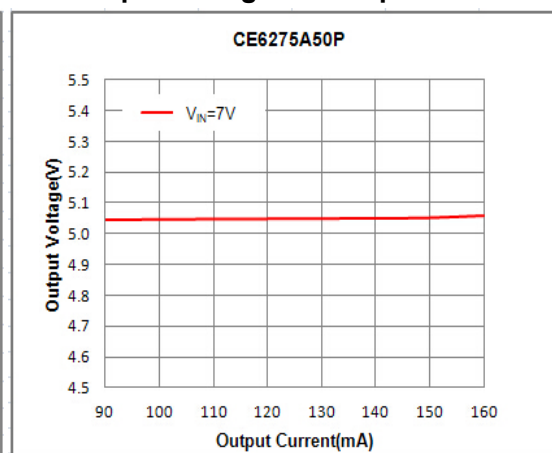


CE6275A50P Typ

Output Voltage Vs. Input Voltage



Output Voltage Vs. Output Current

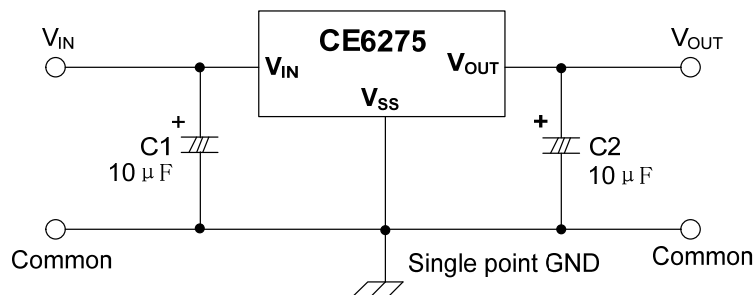


■ Application Circuits

The circuits provided in this section are for reference only.

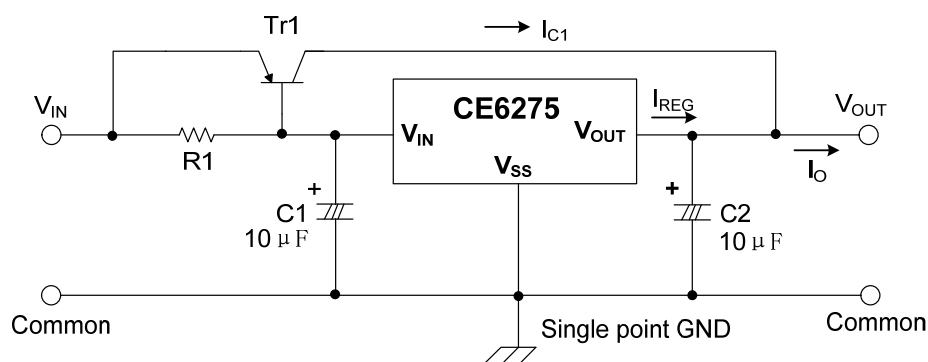
For the purposes of clarity some of the detailed components parameters shall depend on the application.

1. Basic Circuit



Typical Application Circuits

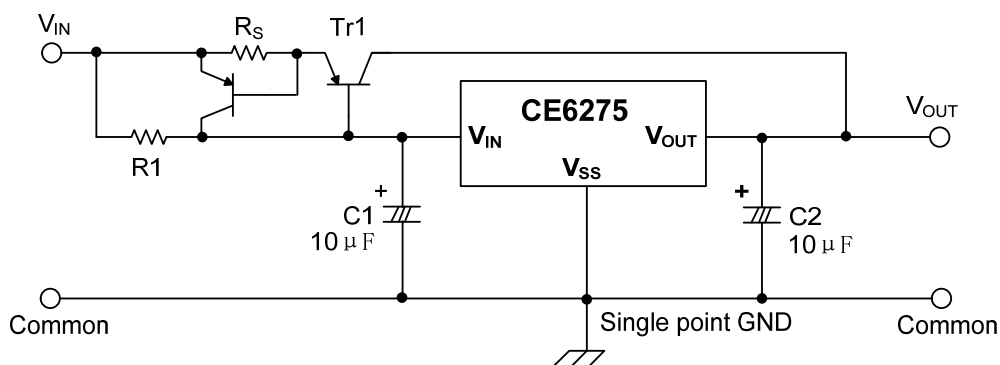
2. High Output Current Positive Voltage Regulator



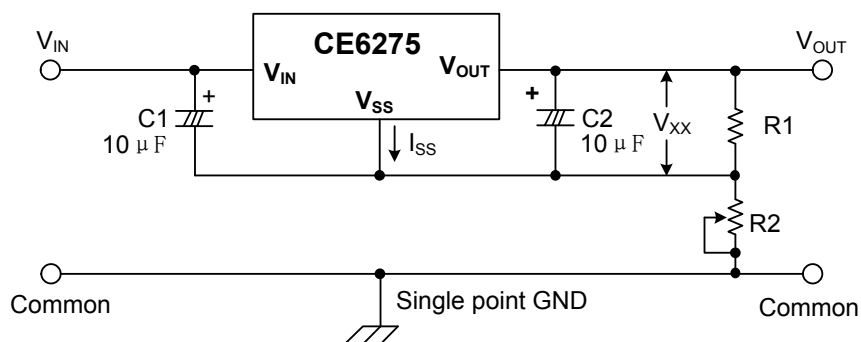
$$R_1 = \frac{V_{BE1}}{I_{REG} - \frac{I_{C1}}{(1+\beta)}}$$

$$I_O = I_{C1} + I_{REG}$$

3. Short-Circuit Protection for Tr1

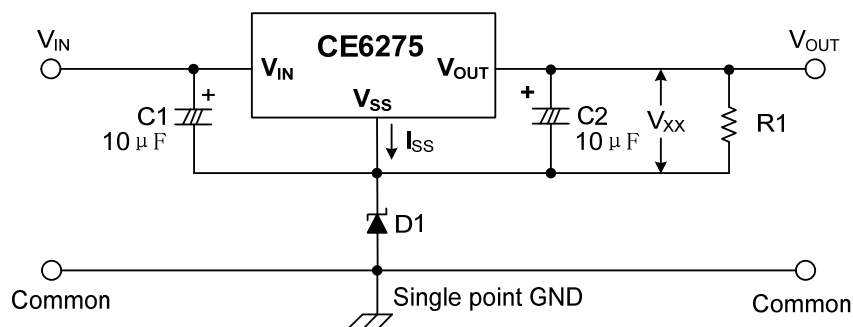


4. Circuit for Increasing Output Voltage



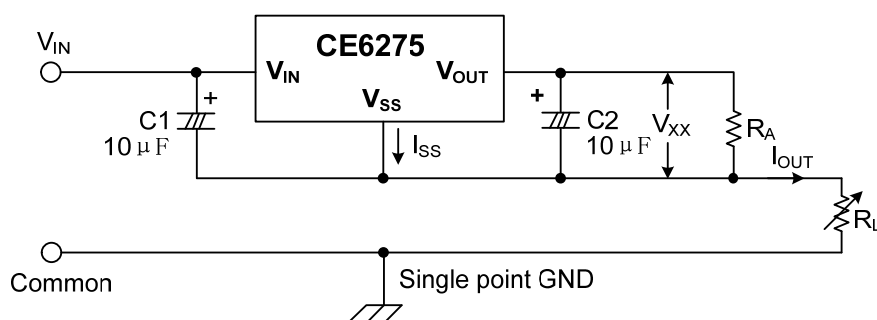
$$V_{OUT} = V_{XX} \left(1 + \frac{R_2}{R_1} \right) + I_{SS} R_2$$

5. Circuit for Increasing Output Voltage



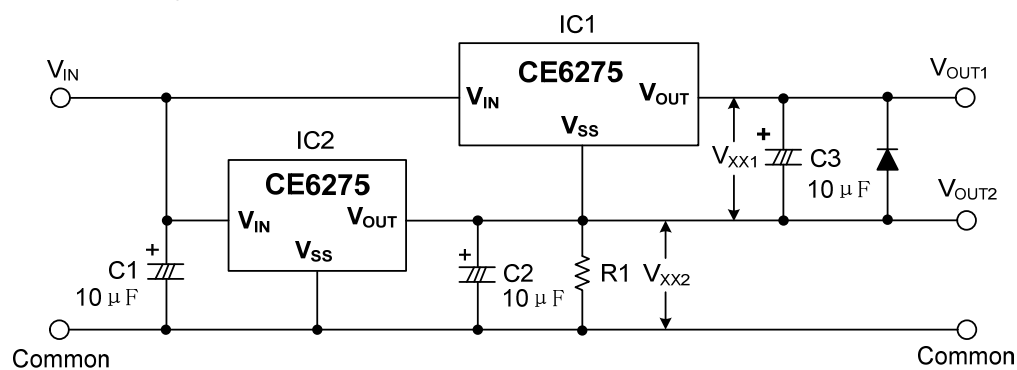
$$V_{OUT} = V_{XX} + V_{D1}$$

6. Constant Current Regulator



$$I_{OUT} = \frac{V_{XX}}{R_A} + I_{SS}$$

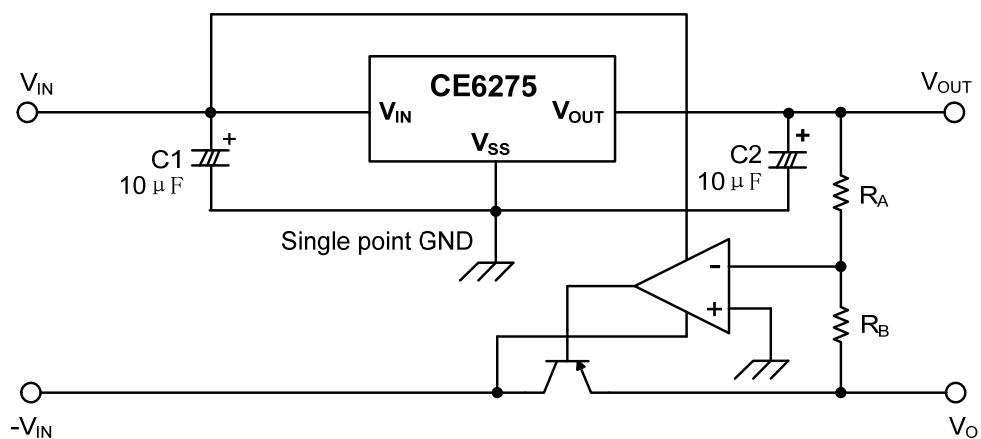
7. Dual Supply



$$V_{OUT1} = V_{XX2} + V_{XX1}$$

$$V_{OUT2} = V_{XX2}$$

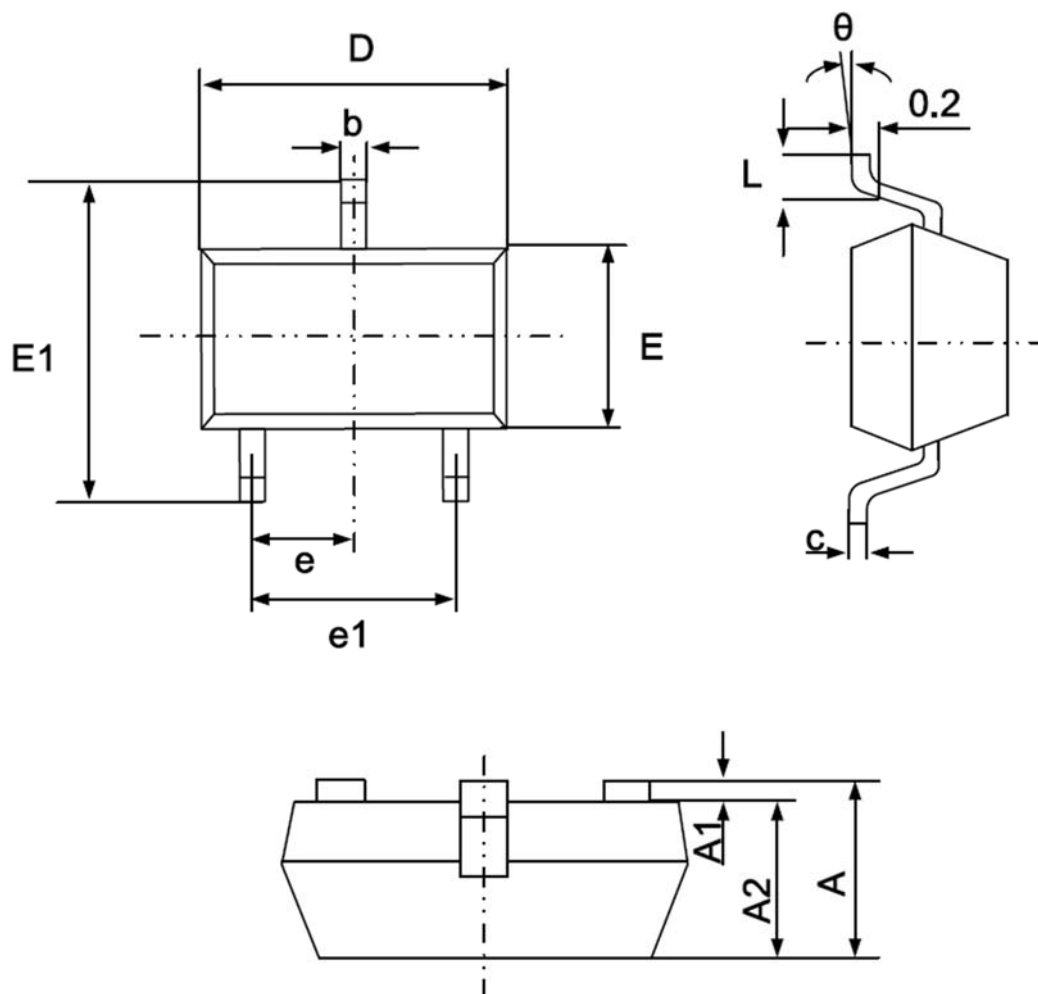
8. Tracking Voltage Regulator



$$-V_O = V_{OUT} \times \frac{R_B}{R_A}$$

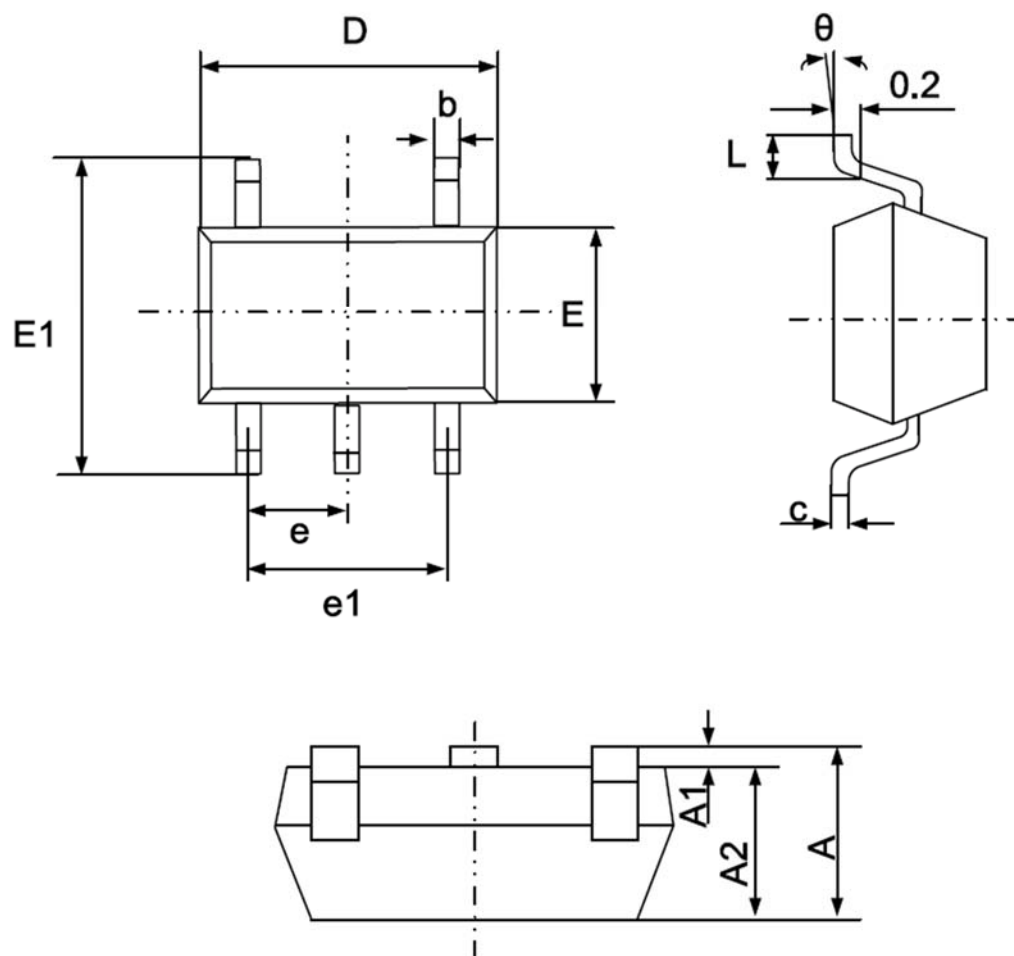
■ PACKAGE INFORMATION

● SOT-23-3 PACKAGE OUTLINE DIMENSIONS



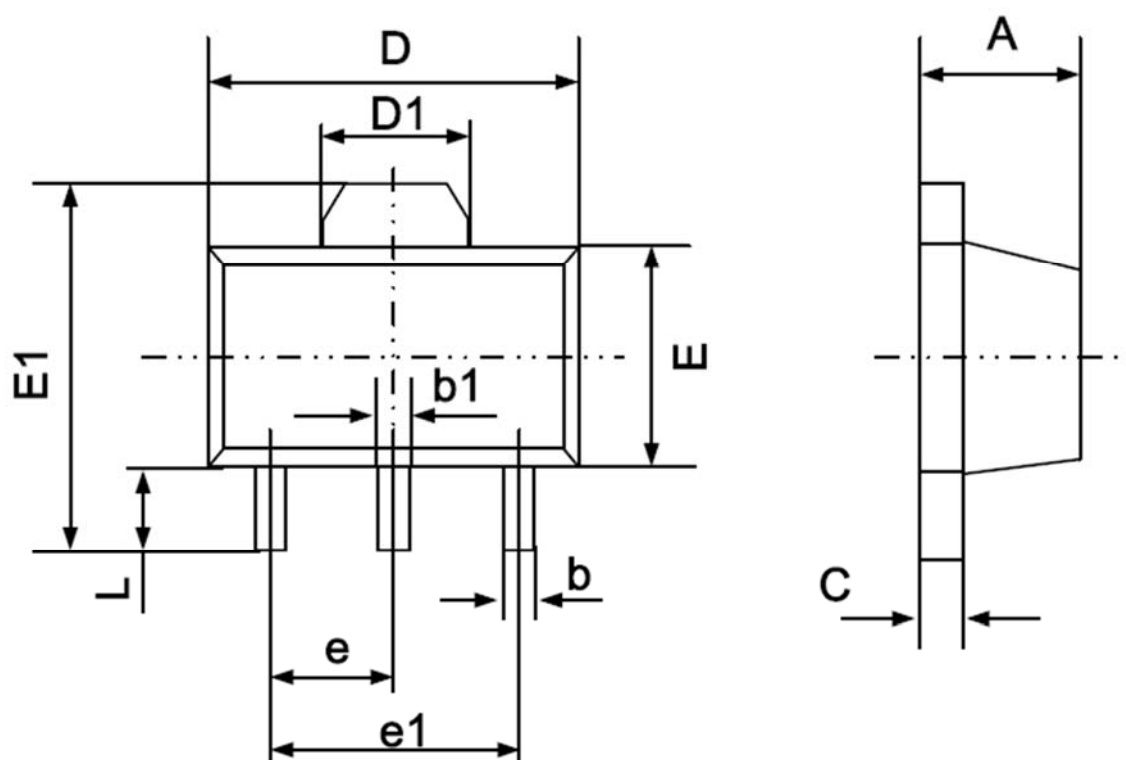
Symbol	Dimensions In Millimeters	
	Min.	Max.
A	1.050	1.250
A1	0.000	0.100
A2	1.050	1.150
b	0.300	0.500
c	0.100	0.200
D	2.820	3.020
E	1.500	1.700
E1	2.650	2.950
e	0.950(BSC)	
e1	1.800	2.000
L	0.300	0.600
θ	0°	8°

- SOT-23-5 PACKAGE OUTLINE DIMENSIONS



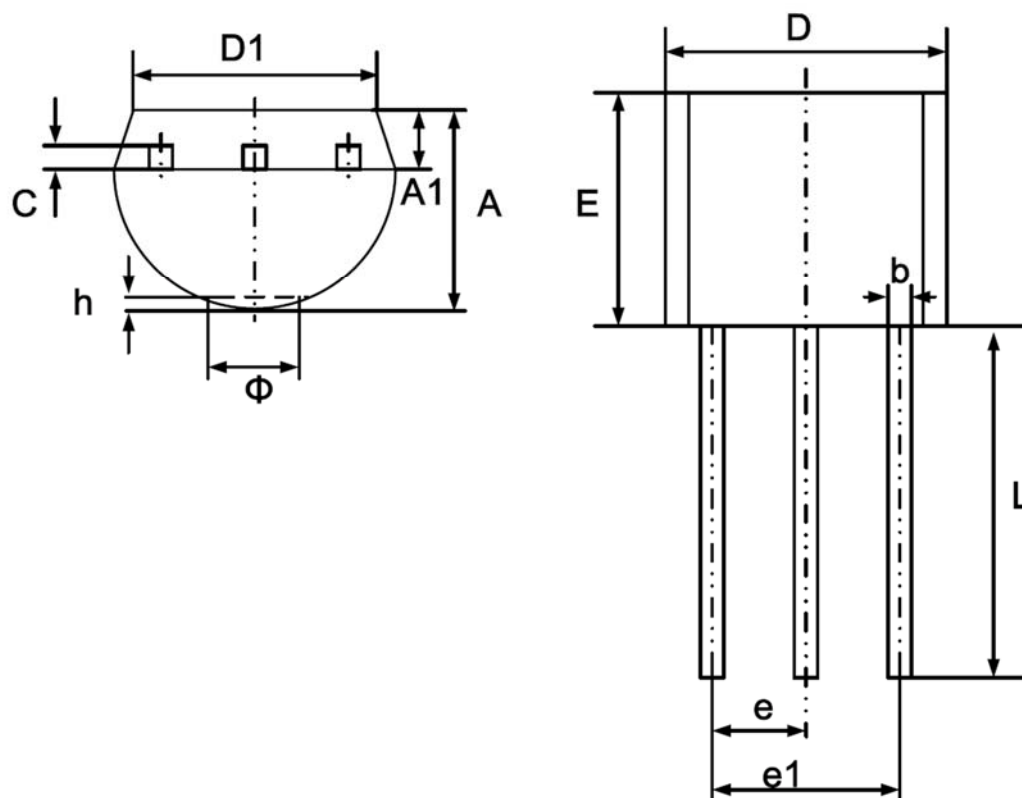
Symbol	Dimensions In Millimeters	
	Min.	Max.
A	1.050	1.250
A1	0.000	0.100
A2	1.050	1.150
b	0.300	0.500
c	0.100	0.200
D	2.820	3.020
E	1.500	1.700
E1	2.650	2.950
e	0.950(BSC)	
e1	1.800	2.000
L	0.300	0.600
θ	0°	8°

- SOT-89-3 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters	
	Min.	Max.
A	1.400	1.600
b	0.320	0.520
b1	0.400	0.580
c	0.350	0.440
D	4.400	4.600
D1	1.550 REF	
E	2.300	2.600
E1	3.940	4.250
e	1.500 TYP	
e1	3.000 TYP	
L	0.900	1.200

- TO-92 PACKAGE OUTLINE DIMENSIONS



Symbol	Dimensions In Millimeters	
	Min.	Max.
A	3.300	3.800
A1	1.100	1.400
b	0.380	0.600
c	0.300	0.500
D	4.400	4.800
D1	3.430	
E	4.300	4.700
e	1.270 TYP	
e1	2.440	2.640
L	13.00	15.00
Φ		1.600
h	0.000	0.380

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